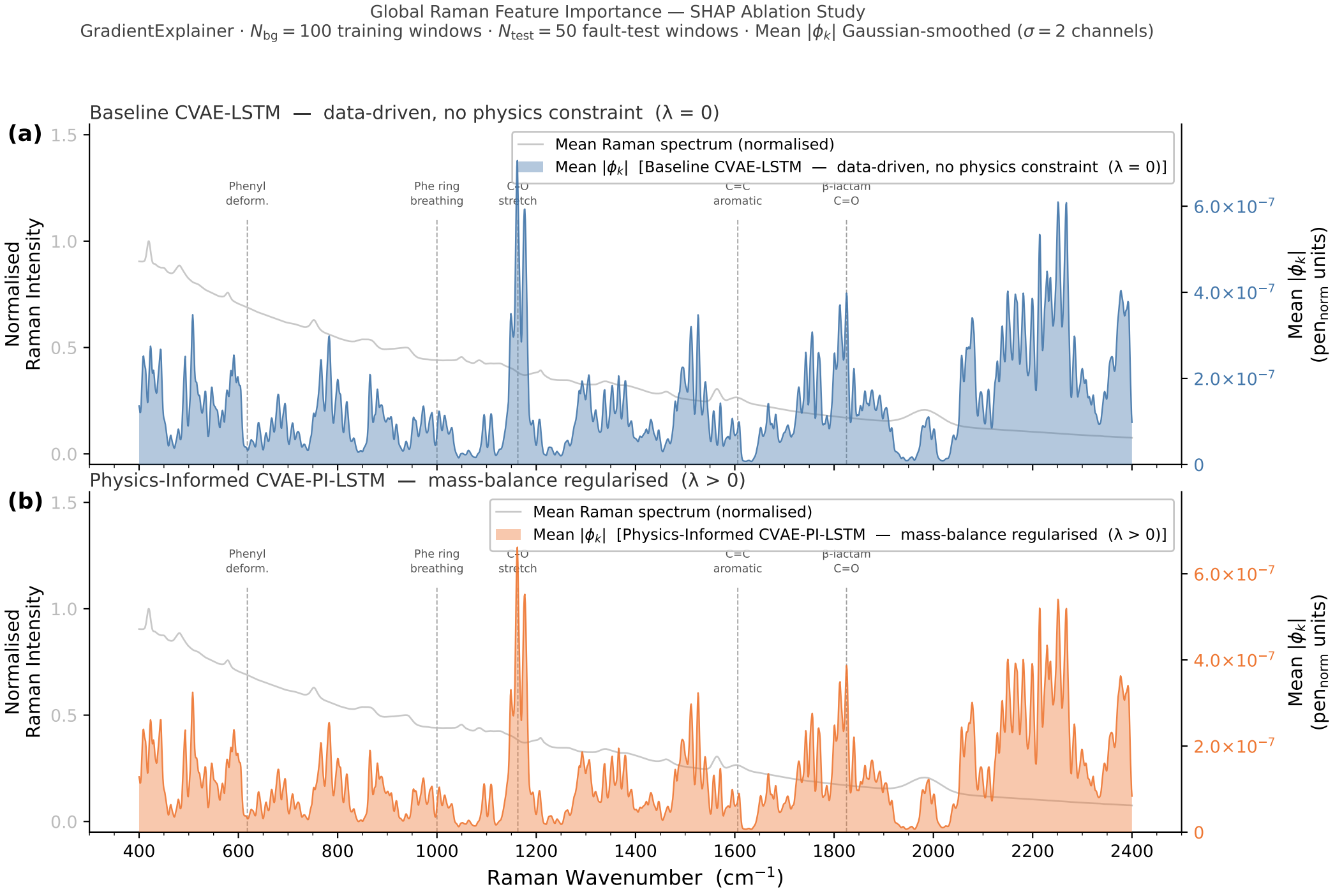

Global Raman Feature Importance — SHAP Ablation Study
GradientExplainer · $N_{bg} = 100$ training windows · $N_{test} = 50$ fault-test windows · Mean $|\phi_k|$ Gaussian-smoothed ($\sigma = 2$ channels)
(a)
Baseline CVAE-LSTM — data-driven, no physics constraint (λ = 0)
Mean Raman spectrum (normalised)
Mean $|\phi_k|$ [Baseline CVAE-LSTM — data-driven, no physics constraint (λ = 0)]
Phenyl deform.
Phe ring breathing
C=O stretch
C=C aromatic
β-lactam C=O
Normalised Raman Intensity
Mean $|\phi_k|$ (pen$_{norm}$ units)
(b)
Physics-Informed CVAE-PI-LSTM — mass-balance regularised (λ > 0)
Mean Raman spectrum (normalised)
Mean $|\phi_k|$ [Physics-Informed CVAE-PI-LSTM — mass-balance regularised (λ > 0)]
Raman Wavenumber (cm^{-1})